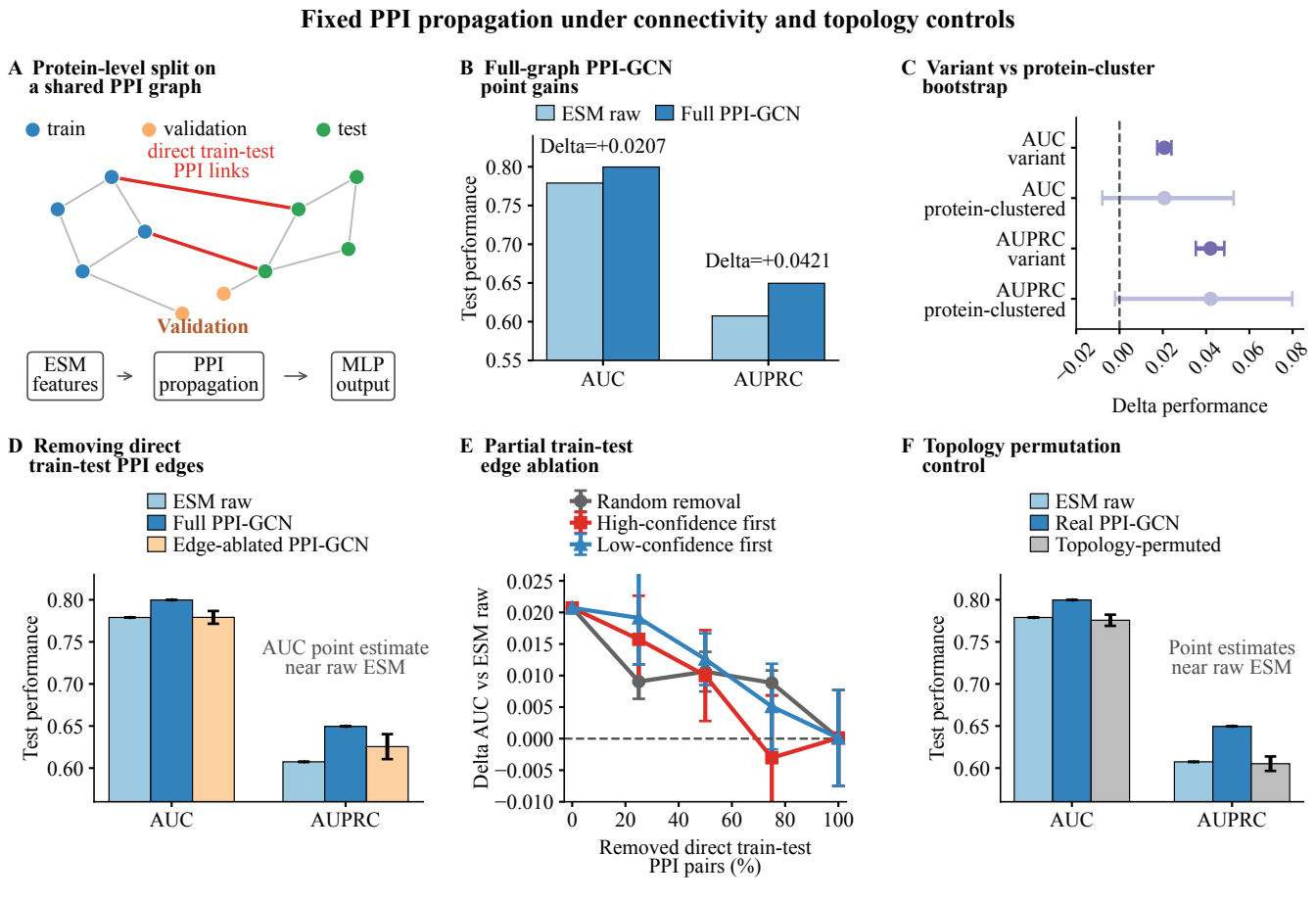

Fixed PPI propagation under connectivity and topology controls
A Protein-level split on a shared PPI graph
train
validation
test
direct train-test PPI links
Validation
ESM features
PPI propagation
MLP output
B Full-graph PPI-GCN point gains
ESM raw
Full PPI-GCN
Delta=+0.0207
Delta=+0.0421
Test performance
0.80
0.75
0.70
0.65
0.60
0.55
AUC
AUPRC
C Variant vs protein-cluster bootstrap
AUC variant
AUC protein-clustered
AUPRC variant
AUPRC protein-clustered
−0.02
0.00
0.02
0.04
0.06
0.08
Delta performance
D Removing direct train-test PPI edges
ESM raw
Full PPI-GCN
Edge-ablated PPI-GCN
Test performance
0.80
0.75
0.70
0.65
0.60
AUC point estimate near raw ESM
AUC
AUPRC
E Partial train-test edge ablation
Random removal
High-confidence first
Low-confidence first
Delta AUC vs ESM raw
0.025
0.020
0.015
0.010
0.005
0.000
−0.005
−0.010
0
20
40
60
80
100
Removed direct train-test PPI pairs (%)
F Topology permutation control
ESM raw
Real PPI-GCN
Topology-permuted
Test performance
0.80
0.75
0.70
0.65
0.60
Point estimates near raw ESM
AUC
AUPRC